

Muhammad Salman Siddiq

Islamabad, Pakistan

✉ salmansiddiq566@gmail.com [in linkedin.com/in/salmansi](https://www.linkedin.com/in/salmansi) github.com/salmansi ☎ +92 331 999 4652

Research Interests

Conversational AI and multi-turn retrieval-augmented generation; agentic search and reasoning; reinforcement learning for LLM post-training; personalization and memory in dialogue; grounded and safe generation for domain applications such as health.

Education

National University of Sciences and Technology (NUST)

Bachelor of Engineering in Software Engineering

Islamabad, Pakistan

Sep 2022 – May 2026

- **CGPA:** 3.54 / 4.00. Dean's List in 7 of 8 semesters.
- **Relevant Coursework:** Machine Learning, Deep Learning, Algorithms, Data Structures, Operating Systems, Cloud Computing.

Research Experience

Thompson Rivers University (TRU), Mamatjan Lab

Research Intern, Mitacs Globalink Research Internship

Kamloops, BC, Canada

Jun 2026 – Aug 2026

- **Multi-turn conversational system:** Designed and built a **multi-agent LLM system** of seven specialized agents, orchestrated with **LangGraph**, delivering source-grounded stroke-recovery education to patients and family caregivers, with a central orchestrator handling intent routing, retrieval, and response composition.
- **Conversational RAG and memory:** Built a RAG pipeline over a curated clinical knowledge base and a **two-layer memory** that separates dynamic user context and preferences from persistent medical facts, improving grounding and consistency across turns.
- **Benchmark and evaluation:** Constructed a **109-question** stroke-caregiver QA benchmark and an automated harness scoring accuracy, source grounding, and safety, with an ablation study isolating each component's contribution.
- **Safety:** Added a guardrail layer to validate and filter model outputs before delivery to the user.

Adaptive Web Interaction: Leveraging RL for Web Action Support

Undergraduate Researcher (Final Year Project)

NUST, Islamabad

Sep 2025 – May 2026

- **Agentic Web Navigator:** Built an autonomous agent that performs user-defined tasks on live websites through an **Observe-Plan-Act-Review** loop, deployed as a Chrome extension that reads the DOM for state observation and executes actions.
- **Distillation Pipeline:** Engineered a **Playwright**-based pipeline that records expert LLM trajectories from real browser interactions, used to distill reasoning from large models into **Small Language Models** via supervised fine-tuning.
- **RL Post-Training:** Trained **Qwen3 1.7B** with SFT followed by **PPO** and **GRPO**, using an **LLM-as-a-Judge** reward to semantically validate task outcomes, and evaluated on a 200-task set filtered from InSTA-150k.

Humanoid RAG Chatbot Avatar

Deep Learning Research Intern

TUKL Deep Learning Lab, NUST

Jun 2024 – Aug 2024

- Developed a humanoid speaking avatar answering user questions in real time, combining **Retrieval-Augmented Generation** with immersive 3D interaction and open-source STT/TTS models (95% accuracy, sub-5 second responses).
- Engineered the LLM-powered RAG pipeline and extended the knowledge base with curated data and numerical reasoning for more precise responses.

Industry Experience

Antematter

GenAI + Backend Engineer (FastAPI, LangChain, PostgreSQL)

Islamabad, Pakistan (Hybrid)

Oct 2025 – Mar 2026

- Built the backend and **AI billing engine** of a production legal billing platform from scratch, classifying emails, inferring client-matter relationships, and generating billable entries by combining LLM reasoning with hierarchical database lookups.
- Designed an **enrichment layer** grounding LLM inference through **hierarchical traversal**, narrowing thousands of legal matters to a few dozen candidates within token limits.
- Developed an **inference layer** with swappable providers, dynamic rule injection, Chain-of-Thought reasoning, and strict JSON outputs for high-precision classification.

- Shipped OAuth integrations for 5 platforms, async webhook and file pipelines with retry logic, and end-to-end observability, with zero downtime since release.

Antematter

Islamabad, Pakistan

GenAI + Backend Intern (Python, FastAPI, Google ADK, RAG)

Jul 2025 – Sep 2025

- Built a **data analytics copilot** using Google ADK + LiteLLM, turning natural-language questions into SQL/Python executed on relational databases with sub-60s latency.
- Designed a **hierarchical multi-agent architecture** (sequential, parallel, loop) with query rewriting, intent routing, schema retrieval, and column pruning to reduce context noise.
- Developed a **self-healing SQL generation pipeline** using ReAct with writer-reviewer-escalation feedback loops, reaching reliable execution within 5 iterations.

MadeIT

Islamabad, Pakistan (Remote)

Machine Learning Intern (Python, LangChain, LLMs)

Sep 2024 – Dec 2024

- Developed an LLM-powered natural-language search system for e-commerce using **LangChain** and Pydantic, returning relevant results for vague queries such as “best camera phones.”
- Scraped and curated a mobile-phone dataset (Pandas, LLM-assisted parsing) and built a weighted multi-criteria ranking model over specs such as CPU, RAM, and refresh rate.

Selected Projects

News Search Engine | Django, NLTK, Python

- Implemented a news search engine over 100K+ articles with custom parsing, inverted indices, and TF-IDF ranking, serving queries in under 100ms with dynamic ingestion.

ZeroHunger Sync | Node.js, PostgreSQL, Prisma, Docker, AWS EC2

- Built a food rescue platform matching providers with NGOs, with location-aware matching, atomic donation-claim workflows, RBAC, and CI/CD deployment on AWS.

Technical Skills

ML / LLMs: PyTorch, HuggingFace, Unsloth, LoRA/QLoRA, SFT, PPO, GRPO, Weights & Biases, Ollama

Agents / RAG: LangGraph, LangChain, Google ADK, vector databases (ChromaDB), LLM-as-a-judge evaluation

Languages: Python, C++, SQL, JavaScript, TypeScript

Systems: FastAPI, Django, PostgreSQL, MongoDB, Docker, GCP, AWS, Git, Linux

Honors and Open Source

2nd Place in North Pakistan, Cambridge A-Level Examinations: ranked 2nd among 100,000+ students on cumulative Physics, Mathematics, and Chemistry results.

NUST Entrance Examination: merit position 189 out of ~50,000 candidates nationwide.

HEC National Skill Competency Test: ranked in the top 1% nationally.

Mitacs Globalink Research Internship (2026): competitively selected, fully funded research internship in Canada.

4th Place, Kibo Robot Programming Challenge 2025 (Team AstroNUTS): 4th of 51 international teams; built a YOLOv11 vision pipeline with ArUco preprocessing for the Astrobeer robot.

Open Source: contributor to **omegaUp** (300+ stars), an open-source competitive programming platform (PR 8040, 8045, 8115).